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Comparing model predictions for ecosystem-based management[1]

Nis S. Jacobsen, Timothy E. Essington, and Ken H. Andersen

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Abstract: Ecosystem modeling is becoming an integral part of fisheries management, but there is a need to identify differences between predictions derived from models employed for scientific and management purposes. Here, we compared two models: a biomass-based food-web model (Ecopath with Ecosim (EwE)) and a size-structured fish community model. The models were compared with respect to predicted ecological consequences of fishing to identify commonalities and differences in model predictions for the California Current fish community. We compared the models regarding direct and indirect responses to fishing on one or more species. The size-based model predicted a higher fishing mortality needed to reach maximum sustainable yield than EwE for most species. The size-based model also predicted stronger top-down effects of predator removals than EwE. In contrast, EwE predicted stronger bottom-up effects of forage fisheries removal. In both cases, the differences are due to the presumed degree of trophic overlap between juveniles of large-bodied fish and adult stages of forage fish. These differences highlight how each model's emphasis on distinct details of ecological processes affects its predictions, underscoring the importance of incorporating knowledge of model assumptions and limitation, possibly through using model ensembles, when providing model-based scientific advice to policy makers.

Résumé : Si la modélisation des écosystèmes devient une partie intégrante de la gestion des pêches, il est nécessaire de faire la distinction entre les prédictions obtenues de modèles utilisés à des fins scientifiques et de gestion. Nous comparons deux modèles, à savoir : un modèle de chaîne trophique basé sur la biomasse (Ecopath with Ecosim (EwE)) et un modèle de communauté de poissons structurée selon la taille. Les modèles ont été comparés en ce qui concerne les prédictions qu'ils produisent quant aux conséquences écologiques de la pêche afin de cerner les points communs et les différences entre ces prédictions pour la communauté de poissons du courant de la Californie. Nous avons comparé les modèles en ce qui a trait aux réactions directes et indirectes à la pêche d'une ou de plusieurs espèces. Le modèle basé sur la taille prédit une plus grande mortalité par la pêche requise pour atteindre le rendement équilibré maximum que le modèle EwE pour la plupart des espèces. Le modèle basé sur la taille prédit également des effets descendants plus forts des retraits de prédateurs que le modèle EwE. En revanche, EwE prédit des effets ascendants plus forts du retrait de poissons-fourrage. Dans ces deux cas, les différences découlent du degré présumé de chevauchement trophique entre les juvéniles des poissons à grand corps et les stades adultes des poissons-fourrage. Ces différences font ressortir le fait que l'accent mis par chacun des modèles sur différents détails des processus écologiques a une incidence sur leurs prédictions, soulignant du coup l'importance d'incorporer les connaissances sur les hypothèses et limites des modèles, éventuellement en utilisant des ensembles de modèles, lorsqu'il s'agit de formuler des avis scientifiques basés sur des modèles à l'intention des décideurs. [Traduit par la Rédaction]

Introduction

The ecosystem approach to fisheries has emerged as a consequence of increased attention to the direct and indirect effects of fisheries on marine ecosystems (Pikitch et al. 2004). Implementation of an ecosystem approach to fisheries commonly requires mathematical models to describe the interactions that drive ecosystem dynamics. This reliance on mathematical models has fostered widespread development of models with varying complexity, each relying on different assumptions and methods. These range from descriptive statistical models to deterministic models (Plagányi 2007), but few formal and quantitative comparisons of multispecies models that describe the same trophic and temporal parts of an ecosystem exist (Kaplan et al. 2013; Smith et al. 2015; Woodworth-Jefcoats et al. 2015).

Although a wide variety of models have been used to generate strategic ecosystem advice for fisheries, most of these models can be broadly categorized as either food web based mass balance models or size-based models. Food web based models often include many taxa (also beyond fish but often only consider differences in food web position through life cycle for a subset of taxa). In comparison, size-based models (including individual-based models) resolve the entire life cycle of individuals and associate each individual's size with trophic position. Consequently, they typically include only the most important fish species and only a coarse representation of lower trophic level non-fish taxa, with birds and marine mammals absent (e.g., Blanchard et al. 2014). To examine the importance of these fundamentally different frameworks, assumptions, and conceptualizations of natural systems, we compare a common food web model, Ecopath with Ecosim (EwE) (Christensen and Walters 2004), and a size-spectrum model (Hartvig et al. 2011) as representatives of these two types of modeling approaches. We use an ecosystem with a well-studied existing EwE model, the Northern California Current, as the study area (Field et al. 2006).

The two types of models have formerly been used to investigate similar problems, i.e., trophic cascades (Daskalov 2002; Andersen and Pedersen 2010), balanced harvesting (Garcia et al. 2012;

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N.S. Jacobsen and K.H. Andersen. Centre for Ocean Life, DTU AQUA, Charlottenlund Castle, 2920 Charlottenlund, Denmark.
T.E. Essington. School of Aquatic and Fishery Sciences, University of Washington, Seattle, WA 98105, USA.
Corresponding author: Nis S. Jacobsen (e-mail: nsja@aqua.dtu.dk).

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Jacobsen et al. 2014), and impacts of forage fisheries (Houle et al. 2013; Pikitch et al. 2014); however, the differences or similarities in results between the two types of models have not been examined. Our goal is to quantify the direct and indirect effects of fishing on the ecosystem. Rather than an analysis of which model performs best, our analysis investigates differences among the models to reveal structural differences that have consequences for impact assessment of fished communities.

Many food web biomass models used for long-term strategic assessment of fishing are developed in the software package Ecopath with Ecosim (EwE; Christensen and Walters 2004). Ecopath is a steady-state model in which total mass balance describes fluxes between compartments (species or groups, henceforth referred to as species for simplicity) in an ecosystem, and Ecosim is a dynamical extension used to project the system forward in time. One of the assumptions in these models is that any species can be described in a single compartment. There is the option to split single species into so called "multi-stanza" groups (Christensen and Walters 2004). This split causes differentiated productivity (predefined as the natural mortality at equilibrium), diets, and therefore consumption rates between a defined set of life stages. The flux of biomass between different stanzas is assumed to follow von Bertalanffy growth, adjusted by the input consumption. Multi-stanza groups are not widely used in EwE models due to high computational and data costs but are instead applied to a subset of species of particular management concerns. Because our interest is in comparing a compartment-based model with a structured model, we are using a well-tested EwE model that does not use multi-stanza groups.

Recently, increasing attention has been paid to quantitative description of fish communities using size-structured community models. Marine teleost fish have an extended life history in which they grow from small eggs (around 1 mg) to large adult individuals between 4 and 7 orders of magnitude larger than the offspring. During this life history, they go through several different trophic levels, change diet composition, redirect the amount of energy directed at somatic growth, and at some point start reproducing. Models that resolve this life history are often referred to broadly as size-spectrum models. Different size-spectrum models with different properties exist, e.g., no species or trait distinctions (Benoît and Rochet 2004), species represented with traits (Pope et al. 2006; Andersen and Beyer 2006), specific food web representation (Hall et al. 2006), solved as individual-based models (Shin and Cury 2004), or prey switching (Maury and Poggiale 2013). Currently, size-spectrum models stand out as a theoretical approach to fisheries ecology, as most of the models describe generic fish communities rather than specific ecosystems. The nature of the models makes it possible to investigate consequences of selective fishing and targeting of specific parts of a fish community (Hall et al. 2006; Andersen and Pedersen 2010). Here we expand a generic fish community description to a size-based model of fish species in the Northern California Current by estimating a few parameters related to recruitment and system production, thus placing it between a specific food web representation and a trait-based model. The estimation makes it comparable to the fish community in the Northern California Current EwE model.

A general problem in multispecies models is allowing for the co-existence of many species (Hartvig and Andersen 2013). An important distinction between the two modeling approaches applied here is how competitive exclusion is avoided. EwE solves this problem by adding a predator-dependent functional response, the so-called "vulnerability" (Walters et al. 1997; Plagányi and Butterworth 2004; Ahrens et al. 2012). The vulnerability parameter causes the consumption of a predator to decrease when its biomass increases. This effect prevents the occurrence of large predator–prey fluctuations. Furthermore, EwE uses the static food web from Ecopath limiting interactions between species. The size-based model applied here uses a Beverton–Holt like stock–recruitment function that determines the maximum number of successfully hatched eggs per year. Using a stock–recruitment function in fisheries models is a long-standing tradition (Beverton and Holt 1957; Myers and Cadigan 1993), but the mechanics behind this density-dependent regulation are not well understood. In the application here, it can be understood ecologically as a carrying capacity caused by effects not covered in the model (e.g., spatial habitat limitations). Effectively, the stock–recruitment relationship controls the maximum biomass of a species. Feeding in size-based models is regulated by size-specific interactions and a prey-dependent functional response and thus is not species specific. The size-dependent feeding introduces additional density dependence into the model when individuals are competing over the same resource.

In this paper, we show that the different model formulations can lead to different predictions of the trophic responses to fishing. As a case study, we use the EwE model described in Field et al. (2006) for the California Current ecosystem, but we did not consider environmental forcing functions, because our goal was to describe qualitative similarities and differences in predictions of models in a deterministic framework rather than to compare the ability of each model to precisely describe full ecosystem dynamics. We use the fish community in the EwE model to calibrate the equilibrium species biomasses in the size-structured model to give the two models equal baseline biomasses before fishing. The California Current ecosystem is used as a model ecosystem to (*i*) predict maximum sustainable yield (MSY) and F_{MSY} of individual species, (*ii*) compare indirect effects of fishing on the ecosystem, and (*iii*) explore underlying mechanisms causing the indirect effects. We use these predictions to explain differences between the models.

Model descriptions

Ecopath with Ecosim

To allow for batch simulations and transparency of the Ecosim method, EwE was re-programmed outside the software package. The general set of equations from the EwE user manual was used (Christensen et al. 2005), with the exceptions listed below (see Appendix A for a full description). For parameterization of feeding time and vulnerabilities, standard values were used. We did not include calculations of handling time as the standard parameter set prescribes that predators can increase prey handling times without limit (Christensen et al. 2005). The Field et al. (2006) parameterization of Ecopath was used to initialize the simulations.

In our calculations, we used a reformulated version of the EwE consumption equation from Gaichas et al. (2012). The Ecosim consumption, $\frac{Q}{B}$, is formulated as

$$(1)\qquad (Q/B_i)_{j,t} = \frac{\left(\frac{Q}{B_j}\right)^* v_{ij} \mathrm{DC}_{ij} T_j T_i \frac{B_i}{B_i^*}}{(v_{ij} - 1)\left(1 + \frac{T_i}{2}\right) + T_j \frac{B_j}{B_j^*}}$$

where subscripts j and i refer to predators and prey, respectively; $\frac{Q^*}{B}$ and B^* indicate values passed on from Ecopath describing consumption per biomass and biomass, respectively; v is vulnerability; DC is the diet matrix; and T is relative feeding time. The formulation specifies that the $\frac{Q}{B}$ of a predator on a prey will decrease once predator biomass increases and is, therefore, a predator-dependent functional response.

The relative feeding time at time T is a dimensionless number calculated with a monthly time step in EwE (EwE source code; Gaichas et al. 2012):

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$$(2)\qquad T_t = 0.1 + 0.9T_{t-1}(1 - A) + \frac{A\left(\frac{Q}{B}\right)^*}{\left(\frac{Q}{B}\right)_{t-1}} \quad T \le T_{\max}$$

where A is a feeding time adjustment constant (Walters and Martell 2004) and $T_{\max}$ is the maximum relative feeding time, with a default of 2. Subscript $t-1$ refers to the previous time step.

Size-spectrum model

Size-spectrum models build on principles of fish communities structured by size. The full model description with equations and parameters are described in Appendix A. The basic principles of the adopted size-spectrum model were developed in Andersen and Beyer (2006) and have been applied to theoretical fish communities (Andersen and Pedersen 2010; Hartvig et al. 2011). The dynamics governing the model is given by the McKendric – Von Foerster equation:

$$(3)\qquad \frac{\partial N_i(w)}{\partial t} + \frac{\partial g_i(w)N_i(w)}{\partial w} = -\mu_i(w)N_i(w)$$

Growth (g) and natural mortality (μ) shape the size spectrum (N) of each individual species i with individual weight w. In the model, individuals grow until their maximum asymptotic size (W_∞). The predator–prey interaction between individuals is described by the rule "big eat small", formulated by a log-normal size preference (eq. A11). The food web is therefore determined by size difference between predators and prey and not, as in EwE, by a species-specific interaction matrix. The predator–prey interactions are used to calculate mortality (μ) and encountered food (E). How much food an individual encounters depends on the food available in the community spectrum and a search rate that scales with individual size (eq. A12).

How much of the encountered food, $E(w)$, is available for consumption, f, depends on a Holling type II functional response:

$$f(w) = \frac{E(w)}{E(w) + \mathrm{hw}^n}$$

where hw^n is the maximum consumption rate. In contrast to the functional response in EwE, the functional response in the size-based model is exclusively prey dependent. The consumed food is assimilated and used for standard metabolism and reproduction (eqs. A14–A15 and A17–A18). The remaining energy is used for growth, $g(w)$ (eq. A16).

Density dependence is imposed with a Beverton–Holt type relationship with a species-specific maximum recruitment parameter, $R_{\max}$, that determines the maximum recruitment flux (eggs·year^{-1}·km^{-2}). We use this parameter to calibrate the biomass of each species in the size-structured model to the biomass of that species in EwE (Appendix B).

Calibration of the size-based model

The size-structured model describes an interacting fish community; therefore, we included the 30 fish species from the Northern California Current ecosystem Ecopath model into the size-based model described by their asymptotic weight. We use the asymptotic weight for comparison in small, medium, and large size classes in both models. Some fish species in the Northern California Current EwE are pooled in groups (e.g., forage fish and small flatfish); here we refer to them as "species" for simplicity and use an average W_∞ (see Appendix B, Table B1) . The asymptotic weights of species in the California Current Ecopath model range between 13 g and 80 kg. Trophic level, as calculated from Ecopath, increases with asymptotic size, but with some variation between species (Fig. 1A). Invertebrates that may be important as prey species (i.e., zooplankton, squid, and crabs in EwE) are included in the size-based model as a dynamic background spectrum that spans from 0.001 g to 50 g (eq. A22).

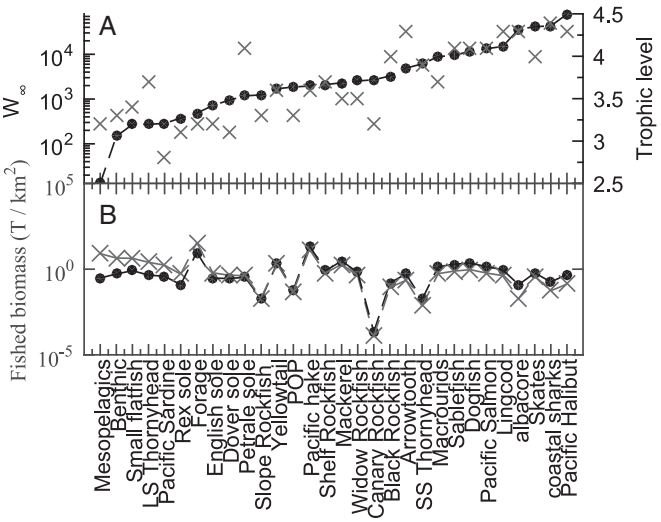


Fig. 1. (A) Solid circles are the asymptotic size (W_∞) of individual species in the size-based model, and gray crosses are the trophic levels of species from Ecopath. (B) Equilibrium biomasses (tonnes·km^{-2}) in the size-based model (solid circles) and in Ecopath with Ecosim (gray crosses).

The fish included in the models represent a wide range of species and functional groups, including flat fish, pelagics, benthic fish, and elasmobranchs. The most speciose group is the rockfishes, which consist of 33% of the number of species modeled, although their biomass is less than 10% of the total biomass.

To facilitate comparison between the two models, we use species biomass as a common metric. The goal of the calibration was to obtain equal biomasses of individual species with no fishing in the models (Fig. 1B). For details on the calibration method, see Appendix B. Species growth rates and asymptotic sizes were obtained from the literature when possible (Kimura 1980; Love et al. 2002; McFarlane and King 2006) or alternatively from Fishbase (Froese and Pauly 2010).

To regulate growth rates in the size-based model, the von Bertalanffy growth parameters from species were averaged over the system, and the emerging growth in the size-structured model was regulated by the maximum food uptake parameter, h, such that modeled growth corresponds to the average von Bertalanffy growth (Fig. 2).

Biomass definitions

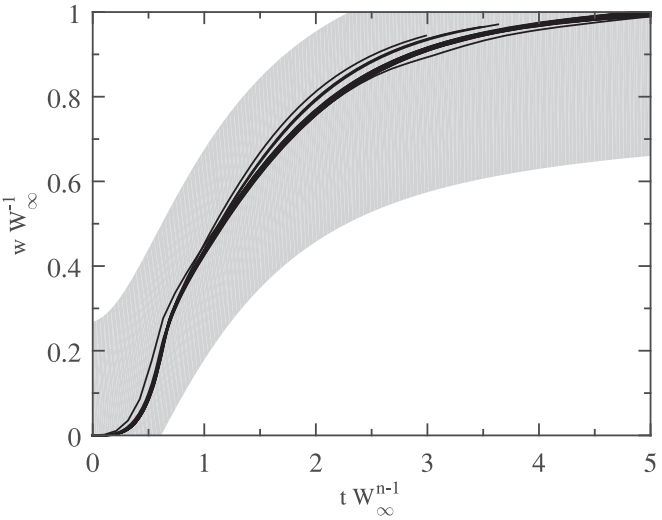
To compare the two models, we need to define what the biomass modeled in EwE represents in the size-structured model. We interpret the biomasses in EwE as the biomass recruited to the fishery. This means that the total biomass of a species is slightly higher in the size-based model than in the EwE due to modeling larval and juvenile fish. Species have different size at recruitment to the fishery and are considered to be recruited with either a knife-edge selection or a trawl-selection (eq. A23; Appendix B, Table B1). The species selectivity curve is assumed to be constant in time. The minimum weight at catch for each species is listed in Table B1.

Fishing scenarios

We distinguish the effects of fishing in the models in two ways: direct effects and indirect effects. The direct effect of fishing is the direct relationship between a fishery and the fished stock, for example, the maximum sustainable yield (MSY) of a stock and what is the fishing mortality needed to reach it (F_{MSY}). In a multispecies context, these results depend on the predation mortality

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Fig. 2. Growth patterns of the 30 species. Black lines represent growth from the size-based model; gray area is the mean of von Bertalanffy growth rates ± 1 standard deviation. The *y* axis is normalized by the maximum asymptotic size (W_∞), and the *x* axis is scaled by the expected life-span ($\propto W_\infty^{(1-n)}$).



and the available food from the rest of the community. For the calculations presented here, the reference points F_{MSY} and MSY are computed without fishing on the remaining community and assume no technical interactions in the fishery.

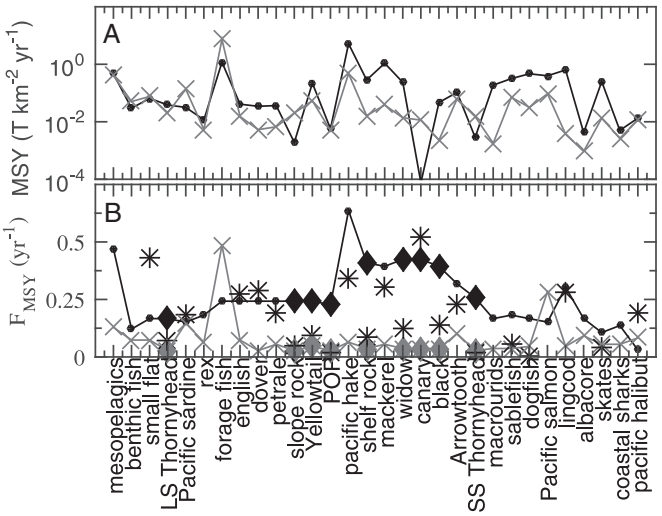
Indirect effects are the changes in the surrounding community when a species is fished. The community responses are a function of direct and indirect releases of predation mortality and changes in food abundance and competition. Here we test the indirect effects by (*i*) fishing part of a community, i.e., small (W_∞ < 1176 g), medium (1241 g ≤ W_∞ ≤ 4903 g), and large (6096 g ≤ W_∞ ≤ 78 492 g) species, divided into three categories with 10 species in each based on their asymptotic size and looking at the biomass responses of the other categories while holding their $F = 0$, and (*ii*) reducing the biomass of one species at a time and measuring the biomass response of the remaining community.

Results

Direct effects of fishing

MSY, when all others are unfished, is similar for some species in the two models, likely owing to their unfished biomasses being similar, whereas others show considerable variation (Fig. 3A). MSY, as we have calculated it, depends on both unfished biomass and growth rate of the individual species. As unfished biomass of a species changes with the surrounding community, the indirect responses of fishing individual species will affect both MSY and F_{MSY}. The fishing mortality leading to MSY, F_{MSY}, differs between the two models (Fig. 3B). Notably, the size-based model generally predicts a higher F_{MSY} than EwE for smaller species (0.2–0.5·year^{-1} for the size-based model; 0.05– 0.2·year^{-1} for EwE). For large species, both models predict $F_{MSY} \approx 0.1$·year^{-1}. The F_{MSY} computations are compared with the F_{MSY} from stock assessments, which we consider the current best guess. The size-based model does not describe low-productivity rockfishes (*Sebastes* spp.) very well (Fig. 3B, diamonds). This is because the growth is averaged in the system, so less productive species are generally overestimated. EwE predicts low F_{MSY} for all species (<0.2·year^{-1}), with forage fish and salmon being higher than others (F_{MSY} = 0.75 and 0.5·year^{-1}, respectively). This causes the F_{MSY} of *Sebastes* spp. to be well estimated by EwE, whereas most other species are underestimated.

Fig. 3. (A) Predicted MSY (tonnes·km^{-2}·year^{-1}) from the size-based model (solid circles) and EwE (gray crosses). The size-based model predicts higher MSY for larger species. (B) F_{MSY} from the size-based model (solid circles), EwE (gray crosses), and stock assessments (black asterisks). Diamonds denote the rockfish genus (*Sebastes* spp., including *Sebastopulos* spp.). The general trend from the size-based model is a flat F_{MSY} slowly declining with asymptotic size. EwE generally predicts fairly low F_{MSY} with the exception of forage fish and Pacific salmon.



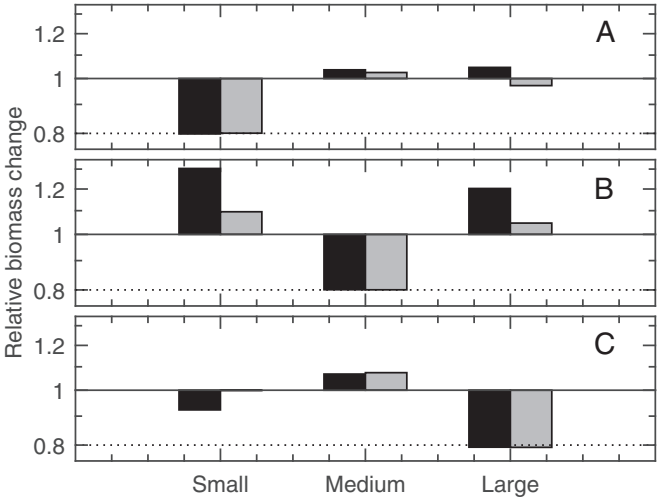
Indirect effects of fishing

To examine fish community effects of targeting certain parts of an ecosystem, we evaluated the indirect effects of fishing small-, medium-, and large-sized species down to 80% of their unfished levels (Fig. 4). In both models, a top-down trophic response is observed in which a release in predation pressure causes increased abundance of prey species (small- and medium-sized; Figs. 4B and 4C, respectively). Interestingly, both models also show the possibility of increased abundance of predator groups when their prey is fished. This is the case in both models when fishing small- and medium-sized species. This effect is caused mainly by reduced competition. Generally, the indirect effects are much smaller than the direct effects (for example, fishing-down the largest predators to 80% of their unfished abundance causes the medium-sized species to increase < 10%). When fishing small species, there is ≈5% biomass increase of intermediate-sized species in both models. However, the effects on large species differ between the models: EwE predicts that large species will suffer from having their prey removed (–5%), whereas the size-based model predicts that large species will increase in abundance (+5%). Fishing intermediate-sized species causes the same response in both models: both small and large species increase. However, the two models predict very different magnitudes of response, with EwE predicting about 10% and 5% increases for small and large species, respectively, and the size-structured model predicting increases of >20% for small species and ≈20% for large species. This result is driven by the predicted response to fishing Pacific hake (*Merluccius productus*), which has relatively high biomass and a diverse diet consisting of multiple trophic levels. The two models also produce divergent predictions about the consequence of fishing large species. In both models, the release in predation pressure causes a 10% increase in medium-sized species, but the effect further down the food web is different. EwE predicts that fishing large species does not influence small ones, whereas the size-based model predicts a trophic cascade, causing small species to decrease ≈8% in abundance.

Both models predict changes to community biomass distributions in response to fishing predator species (Figs. 5 and 6). In EwE, the indirect responses on the species level depends on the struc-

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Fig. 4. Indirect biomass changes when fishing (A) small, (B) medium, or (C) large (see main text) fish species down to 80% of unfished biomass. Black is the size-based model, gray is EwE. Fishing small species has the same impact on medium species in the two models (+), but opposite effects on large species, with the size-based model predicting (+) and EwE predicting (–). Fishing medium-sized species has positive effects on both small (+) and large (+) species in both models. Fishing large species causes similar responses in the two models to the medium species (+), but the size-based model predicts tertiary effects on the small species (–).



ture of predator–prey linkages that comprise the modeled food web, where interaction strengths determines the indirect effects of fishing (Fig. 5A). Even without an explicit food web, the size-based model shows some of the same patterns as in EwE (Fig. 5B), i.e., prey species increase in abundance and there is an increase in abundance of the species close to the asymptotic size of the fished species. The main differences between the two models are seen further away from the fished species, where large species generally decrease in abundance in EwE, and trophic cascades range over more trophic levels in the size-based model (Figs. 5 and 6). Both models show that forage fish and Pacific hake exert the largest response on the surrounding community. Removal of prey causes a decline of their predator species in EwE, but surprisingly, the opposite is evident from the size-based models. For example, when "forage fish" are targeted, their predators increase their biomass (Fig. 7).

Taken together, the most striking difference between the models is the difference in bottom-up effects of depleting lower trophic level species. In EwE, a fishing-induced depletion of a trophic level generally led to a reduction in higher trophic levels, whereas the size-based model predicts an increase in higher trophic levels for all species (Figs. 5 and 6). This difference is caused by explicit size structure of the size-based model, which leads to intra- and inter-specific competition at younger life stages. The competition in those stages induces an integrated positive response throughout the food web when a certain species is fished, here shown with "forage fish" (Fig. 7). The effect is mainly caused by a release of predation pressure (Fig. 7B). The higher survival of younger fish causes an increased flux to mature individuals of larger species, ultimately increasing the biomass of older life stages. For small species, this effect also has positive effect on recruitment, as it is mainly mature fish that are released from predation pressure (Fig. 7A).

Discussion

We have shown differences and commonalities between the two models currently being used for medium- to long-term assessment of the impact of fishing. There are three main structural differences in the models. First, the models define biomass dynamics differently. In EwE, it is assumed that the species biomass dynamics are described in a single compartment, whereas the size-spectrum model resolves each species into a continuum of size classes. Second, density dependence in the models is described in two fundamentally different ways. In EwE, density dependence is imposed as a predator-dependent functional response (vulnerability), whereas the size-spectrum model uses a stock–recruitment function. Third, trophic interactions in EwE depend on assumed species-level linkages among compartments, whereas the size-structured models depend on assumed predator–prey size ratios to produce predator–prey linkages.

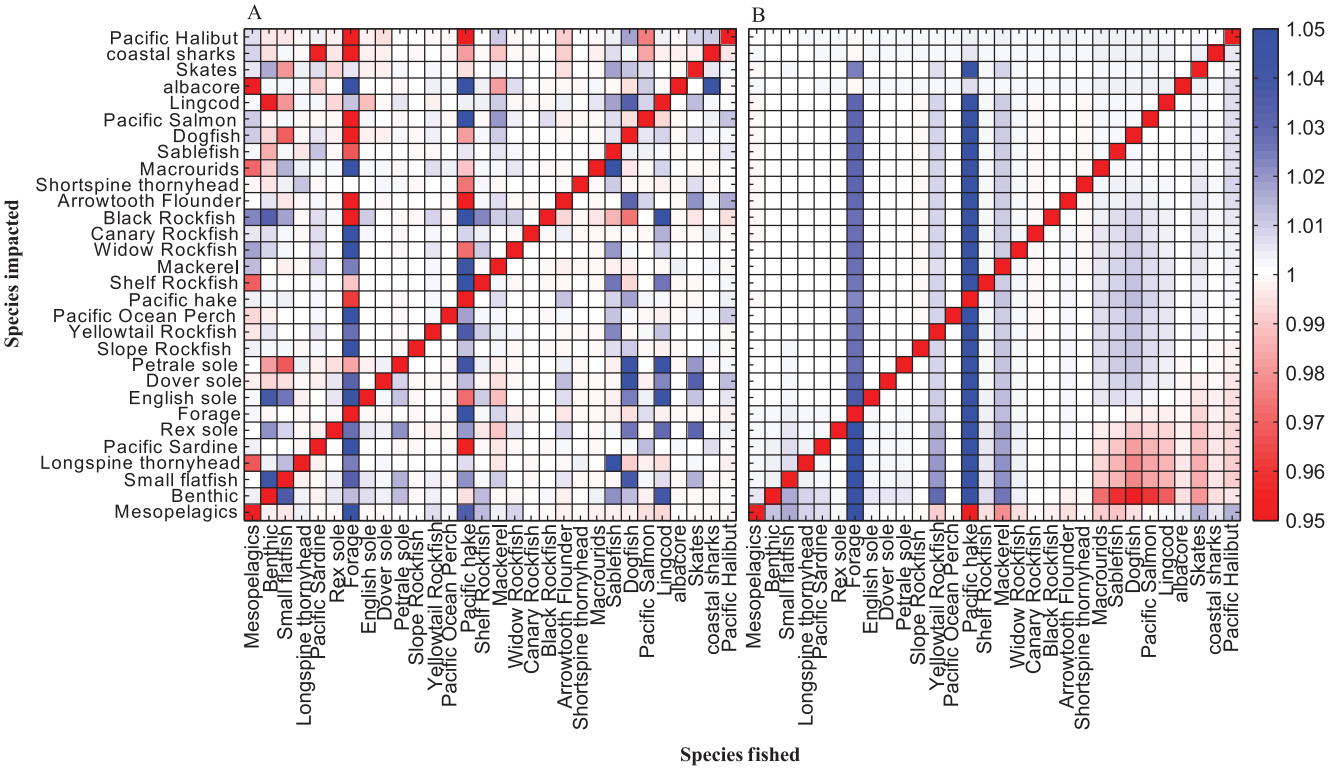
Introducing size-structured predator–prey linkages leads to competition among species at different trophic levels due to ontogenetic diet shifts. In contrast, compartment models do not necessarily represent this linkage, unless models were designed with explicit consideration of ontogenetic diet shifts. There is ample evidence for this type of trophic omnivory in marine ecosystems producing intraguild predation (Polis and Holt 1992; Köster and Möllmann 2000). On the other hand, size-spectrum model predictions might overestimate the degree of resource overlap if there are other factors distinct from size that govern resource partitioning, e.g., differentiated habitats.

As a consequence of the structural assumptions in the size-based model, early life competition is strongly emphasized such that fishing the small-sized fishes can cause a seemingly counter-intuitive increase in large fish abundance via competitive release. These responses are less pronounced in compartment-based models such as EwE in which complex food web interactions are the main drivers of changes in abundance. Consequently, application of compartment models to predict response of fishing small-sized fish (Smith et al. 2011; Pikitch et al. 2014) might not capture the full range of responses of upper trophic level fish that might be competitors at early life stages. From a meta-analysis of Ecopath models, Essington and Munch (2014) found that fishing forage fish could have a variable influence on the predator fish yield because of complexity in food webs induced by intraguild predation at several levels of food webs. Recent analysis of a size-based model indicated that a strong reduction in smaller trophic levels can also lead to a decreased yield from higher trophic levels (Houle et al. 2013), suggesting that there may be a threshold at which depleting forage fisheries to lower levels than presented here will have net negative effects on their predators.

The implications of food web approaches versus the size-structured approach for predicted response to fishing are mostly evident at the species level. An important assumption in EwE is the knowledge of link strengths between each species in the food web, as well as species productivity. The assumption requires a great amount of data and system knowledge and may eventually lead to large uncertainties in model output (Essington 2007). Link strengths between species can be interpreted as a combination of food preference and spatial overlap. The mechanistic approach of the size-based models assumes a homogeneously mixed community in which interactions are only size based. This approach requires much less parameterization and seems to catch the overall patterns well, although presumably at a cost of relying on assumptions that are derived through generalized size-based relationships. However, it does not account for the feeding patterns and productivities of certain species, which is reflected in the general overestimation of rockfishes F_{MSY}. It is possible to add additional complexity to the size-based models. Food webs can be constructed (Hartvig et al. 2011) and individual growth rates can be regulated by the parameter controlling maximum consumption rate (Blanchard et al. 2014). Furthermore, species-specific preferred predator–prey mass ratios and the widths of the food preference function can be estimated (Blanchard et al. 2014; Zhang et al. 2015). For these estimations, it is important to distinguish between the preferred and the realized predator–prey mass ratios (Hartvig et al. 2011), as the realized predator–prey mass ratio de-

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Fig. 5. Matrix of (A) EwE and (B) size-based model showing the relative biomass changes to the community when fishing one species at a time down to 80% of unfished biomass, i.e., the diagonal is the direct effect of fishing, above the diagonal is the indirect biomass changes on the larger species, and below the diagonal is the changes on smaller species. In both models, species close to the diagonal benefit from removal of competitors. Generally, the largest species decline in biomass in EwE, whereas they increase in the size-based model when lower trophic levels are fished. Release of predation pressure causes an increase of lower trophic levels in both models, but there is a tertiary effect in the size-based model causing small species to decline when the largest are removed.



pends on the available prey in the system. The choice of parameters for the prey selection curve does not influence the magnitude of trophic cascades observed in the size-spectrum model, but changes in the preferred predator–prey mass ratio shifts the position of the cascades slightly (Andersen and Pedersen 2010). Some of the differences that arise due to having an unstructured model, e.g., gear selectivity and juvenile stage competition, would become less apparent if more focus was on multi-stanza groups when developing EwE models.

Another difference between the models is that EwE explicitly models food webs from plankton and detritus, through intermediate trophic levels, and up to the highest portions of food webs, whereas the size-based model only describes the fish community. An assumption in the size-based model is that direct and indirect effects can be described primarily by the fish community, although that is not always the case (Mackinson 2013). Smaller food items (i.e., zooplankton and other invertebrates) are modeled as a simplified dynamic background spectrum. EwE allows the inclusion of invertebrate groups that could be important for dynamics or fisheries, i.e., squid or crabs (Kaplan et al. 2013), and can also be calibrated bottom-up with temperature forcing. Full ecosystem models allow us to evaluate both direct effects of fishing and the indirect effects generated throughout the food web.

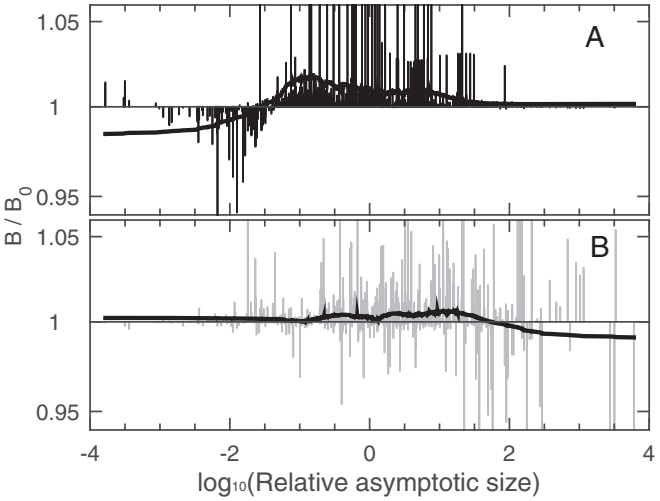
The different approaches used to promote stability in the models (i.e., predator-dependent functional responses versus density-dependent recruitment) do not seem to cause crucial differences in the model outputs that we examined here. In EwE, the strength of the trophic responses shown would be smaller with a smaller vulnerability and, conversely, higher with increased vulnerability (Walters et al. 1997). In the size-based model, changes in maximum recruitment could change the relative magnitude of F_{MSY} between small and large species (Andersen and Beyer 2015). For the indirect effects further away from the impacted species, the influence of the food web in EwE is more apparent. For example, the food web prescribes that 50% of the diet of the arrowtooth flounder (*Atheresthes stomias*) is comprised of Pacific hake (Field et al. 2006). This link causes arrowtooth flounder biomass to decline in EwE when Pacific hake is fished down 20% (Fig. 5A). The lack of species preference in the size-based model causes arrowtooth flounder to choose other available prey items around that size class, whose biomass increase due to lowering of competition caused by the fished Pacific hake (Fig. 5B).

The F_{MSY} provided in this study are reference points calculated one species at a time without fishing the remaining community. Incorporating all fishing mortalities will lead to differentiated F_{MSY} due to changes in natural mortality caused by the biomass levels of predators and competitors (Gislason 1999). Generally, reference points depend on the natural mortality, the productivity of each species, the steepness of the stock–recruit relationship, and the selectivity of the gear. How F_{MSY} is distributed with asymptotic weight in the size-structured model largely corresponds to the patterns in recent calculations from a single-species size-based model (Andersen and Beyer 2015). The F_{MSY} pattern found in EwE can to some degree be attributed to the fact that the whole population is vulnerable to fishing compared with the size-structured model in which mesh sizes protect younger individuals. These results show quantitative and qualitative differences between the two models.

The analysis calls for an evaluation of how multispecies models should be used in practical management. The results here show differences between F_{MSY} and MSY in the models that suggest that these reference points will be estimated differently with imperfect knowledge of selectivity, growth, and interactions, as well

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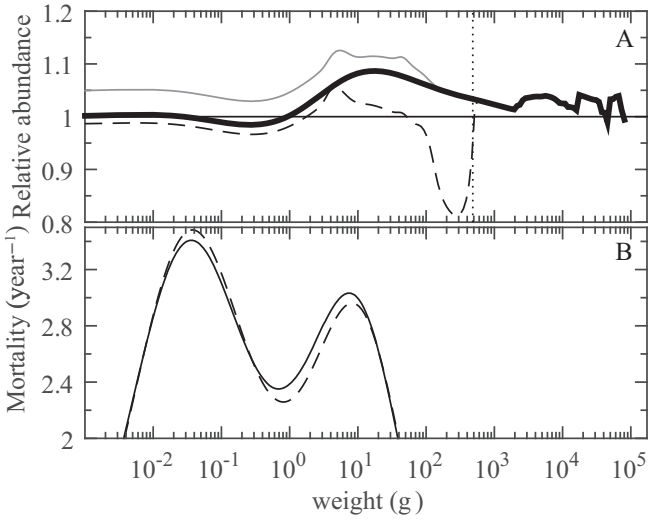
Fig. 6. Pinplot of Fig. 5 showing the relative community changes when removing 20% of each species, i.e., 0 on the *x* axis is the same asymptotic size as the species being removed (fished species not included in figure). A is the size-based model and B is EwE. The black line is a smoothing filter to show general trends. Common for both models is that around the asymptotic size of the removed species, other species increase in abundance. In the size-based model, this causes a cascade down the food web, whereas in EwE, only the prey of the removed species increase, but with lower intensity than in the size-based model. Larger species increase in abundance due to decreased competition in both models. In EwE, the largest species decrease. The smoothing lines show that the main model difference is in the tertiary effects.



different underlying ecological assumptions. Reference points such as F_{MSY} and MSY are often estimated in single-species models and practically used for tactical management. Multispecies models can be run in ensembles to investigate the long-term effect on an ecosystem by applying single-species reference points or alternatively other ecosystem strategies, e.g., Rice and Duplisea 2014. In the case in which an ensemble of multispecies models agrees on the indirect effects of fishing, the predictions can be used to guide ecosystem-based management. When there is broad agreement in predictions that are based on widely different structural assumptions, one has stronger confidence that the predictions are robust because they do not appear sensitive to these assumptions (these assumptions are typically about processes for which there is little available data for validation) (Gårdmark et al. 2013; Smith et al. 2015). Ensemble runs can explore if any species go beneath biomass limits under the desired exploitation pattern or how yields from interacting fish populations can affect one another. Further exploratory analysis can propose alternative management actions to avoid collapses or undesired changes in species compositions. In cases in which models disagree, the underlying effect causing the discrepancy needs to be identified before the model ensemble can be used to guide management. In the case of the Northern California Current, the models presented here were fairly consistent in their prediction of indirect effects on competitors, as well as the increase in prey biomass when predators were fished, despite fundamentally different model formulations. When fundamentally different models predict similar responses to a pressure such as fishing, the confidence in robustness of model projections and structural uncertainty is enhanced.

The size-spectrum model was calibrated to the California Current EwE model and revealed both differences and commonalities. The following main differences between the models were identified: (*i*) EwE predicts lower F_{MSY} than the size-structured model; (*ii*) the trophic cascades have different magnitudes and

Fig. 7. (A) The size spectrum (individual weights) relative to the unfished spectrum when fishing down the "forage fish" species 20% in the size-based model. When the lines are > 1, biomass has increased compared with the unfished situation. The dashed line denotes all fish (including forage fish), the thick black line is the size spectrum of species with W_∞ larger than "forage fish", and the thin gray line denotes the size spectrum of all species excluding "forage fish". The vertical dotted line is W_∞ of "forage fish". Most of the individuals removed from "forage fish" are around their size at maturity, causing individual fish around 1 g of other species to increase in abundance, including those with large W_∞. The change is primarily caused by reduced predation mortality and causes more individuals to survive until adulthood (see panel B). Smaller species also get higher reproductive output (gray line). (B) Individual size predation mortality. The continuous line is the mortality at $F = 0$, the dashed line is when "forage fish" are fished down 20%.



work differently for larger trophic levels; and (*iii*) competition between juvenile stages in the size-spectrum model are important for overall dynamics. MSY predictions between the two models were similar for some species but with considerable uncertainty in other species. The main commonality between the two models was to predict indirect effects of competition and mortality of species close to the one being targeted. We consider the effect of juvenile competition to be a general difference between size-based models and compartment-based models and thus an important topic for future investigation. The pros and cons of different models employed in the ecosystem approach to fisheries call for an integrated approach when doing impact assessments of fisheries on long to medium time scales. We conclude that structural uncertainty in models should be accounted for when evaluating the consequences of fisheries on marine ecosystems. Uncertainty in strategic predictions of fishing scenarios can be investigated by running model ensembles that include different mechanistic assumptions and comparing responses on the ecosystem level. Here we have identified that dynamics of juvenile life stages can cause differences in indirect effects of size-based versus compartment-based models. We also acknowledge link strengths between predators and prey to be an important driver of indirect effects between species. We recommend careful attention to how differences between prey availability and prey preference can shape the conclusions from food-web-based and size-based simulation models.

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Appendix A

Model formulations

Simulations from both models were performed in MATLAB version 2014a.

Ecopath with Ecosim

The first step of a dynamical Ecosim model is the balancing of rates in the static Ecopath with the conservation equation (Polovina 1984):

$$\text{(A1)} \qquad B_i\left(\frac{P}{B}\right)_i \mathrm{EE}_i + \mathrm{BA} = \sum_j B_j\left(\frac{Q}{B}\right)_j \mathrm{DC}_{ij} + C_i$$

The left side of eq. A1 describes the positive terms, where B is the biomass of species i, P is production, EE is ecotrophic efficiency (the amount of mortality not included in the model), and BA is biomass accumulation. The right side describes the loss terms in the equation, where Q is consumption, DC is relative diet composition of prey i of predator j, and C is catch. Typically, biomasses, catches, production per biomass, consumption per biomass, and diet composition are given to Ecopath, which allows for a calculation of

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the ecotrophic efficiency. In principle, any of the parameters or states could be calculated by solving the linear systems.

Ecopath also calculates a gross food conversion efficiency, G_i:

$$G_i = \frac{\frac{P}{B_i}}{\frac{Q}{B_i}} \quad \text{(A2)}$$

The parameterization from Ecopath is then passed on to Ecosim and used in forward projection of the ecosystem. The main dynamic equation in Ecosim is

$$\frac{\partial B_i}{\partial t} = G_i \sum_j Q_{ji} - \sum_j Q_{ij} - (M_{0,i} + F_i + e_i)B_i \quad \text{(A3)}$$

where Q is consumption, M_0 is mortality other than predation, F is fishing mortality, and e is emigration rate. The consumption is originally defined as

$$Q_{ij} = \frac{a_{ij}v_{ij}B_iB_jT_iT_jS_{ij}M_{ij}/D_j}{v_{ij} + v_{ij}T_iM_{ij} + a_{ij}M_{ij}B_jS_{ij}T_j/D_j} \quad \text{(A4)}$$

where a is search rate, v is vulnerability, T is relative foraging time, S is a seasonal forcing effect, M is mediating forcing effects, and D is effects on handling time to maximum consumption rate. We have used a reformulated equation for consumption. See eq. 1 of the main text.

The dynamics of detritus groups are modeled as the sum of egestion and the biomass lost to other mortality (M_0). In addition, there is one detritus group, fisheries offal, whose dynamics depend on the discards of the fishery. For the California Current EwE model, egestion is set to zero, so only losses from natural mortality are directed to detritus by the function

$$P_{\text{det},i} = M_{0,j}B_j - E_i \quad \text{(A5)}$$

where P_{det} is the detritus production group and E is the export.

Nonconsumer dynamics

The California Current EwE model includes two types of nonconsuming organisms, one primary producer and three detritus groups. As these groups do not consume biomass, their dynamics differ from those of other organisms. To include the effects of competition for nutrients and light shading (Walters et al. 1997), the consumption term in eq. A2 for primary producers is substituted with a production term, defined as

$$P_{\text{pp}} = r\frac{B_{\text{pp}}}{1 + B_{\text{pp}}h} \quad \text{(A6)}$$

where r is the maximum $\frac{P}{B}$ that the primary producer can have, and h is defined as

$$h = \frac{\frac{r}{\left(\frac{P}{B}\right)^*} - 1}{B^*} \quad \text{(A7)}$$

causing the primary production to have a saturating relationship. The asterisk (*) indicates that the value is passed from Ecopath to Ecosim.

The applied model was verified against similar model runs in the EwE software to verify that the dynamics were similar, and although not yielding the exact same results, the calculated rates (Q/B, P/Q, and T) were exactly the same. For a few species, there were initial differences in P/B.

Size-based model

We use a version of the model developed in Hartvig et al. (2011). The general model is based on a few assumptions: (*i*) big fish eat smaller fish; (*ii*) species can be described by their individual weight (w) and the trait asymptotic size, W_∞; and (*iii*) the model is mainly driven by individual-level processes such as predation, growth, and reproduction.

In contrast to the approach of Hartvig et al. (2011) and other food web approaches such as that of Hall et al. (2006), we assume that there is incomplete knowledge of the food web. This causes predator–prey overlap to be entirely dependent on individual size, following the first main assumption.

The model framework is well documented in the literature (Andersen and Beyer 2006; Andersen and Pedersen 2010; Hartvig et al. 2011; Jacobsen et al. 2014), but as the goal of this paper was model comparison, all equations are listed in the text that follows and parameters are listed in Table A1.

The dynamics in the model are described by the McKendrick – Von Foerster conservation equation:

$$\frac{\partial N_i}{\partial t} + \frac{\partial g_iN_i}{\partial w} = -\mu_iN_i \quad \text{(A8)}$$

where N_i is the size spectrum of species i, t is time, g is growth, and w is individual weight. N has the unit no.·w^{-1}·vol^{-1} and is thus a density function, which can also be denoted the "number size spectrum". Similarly, the biomass density spectrum can be calculated as $B_i(w) = wN_i$, having the unit vol^{-1}. The integral of the biomass density function gives the actual biomass in the model calculated as

$$B_{\log,i} = \int_{w_0}^{W_\infty} wN_i\,\mathrm{d}w \quad \text{(A9)}$$

which is the species biomass used to calculate Fig. 2 (main text).

Growth comes exclusively from feeding on the community spectrum, which is the sum of the fish size spectrum and a background spectrum, N_R:

$$N_c = \sum_i N_i(w) + N_R(w) \quad \text{(A10)}$$

Feeding on the spectrum is selected using log-normal preference (Andersen and Ursin 1977):

$$\phi\left(\frac{w_p}{w}\right) = \exp\left\{\frac{-\left[\ln\left(\frac{w}{w_p\beta}\right)\right]^2}{2\sigma^2}\right\} \quad \text{(A11)}$$

where w_p is the size of the prey, β is the preferred predator–prey mass ratio, and σ is the width of the preference curve. The amount of biomass that a predator encounters is described by the encounter rate:

$$E(w) = \gamma w^q \int_0^{\max(W_\infty)} N_c(w)\phi\left(\frac{w_p}{w}\right)w\mathrm{d}w \quad \text{(A12)}$$

Besides being limited by the food available from the spectrum, food uptake is controlled by a Holling type II functional response, here denoted the "feeding level":

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Table A1. Parameters used in the size-based model.

Parameter	Value	Definition	Reference(s)
β	100	Preferred predator–prey mass ratio	Andersen and Ursin 1977; Jennings et al. 2001
σ	1.2	Width of selection curve	Andersen and Pedersen 2010
γ	1.33×10^{-6} g^{-q} $year^{-1}$ km^{-2}	Search rate	Hartvig et al. 2011
q	0.8	Search volume exponent	Andersen and Beyer 2006
h	15 g^{1-n} $year^{-1}$	Maximum consumption factor	
n	0.75	Maximum consumption exponent	Andersen and Pedersen 2010
η_m	0.25	Size at maturation relative to W_∞	Beverton 1992
α	0.6	Assimilation efficiency	Kitchell et al. 1977
k_s	2.4 g^{1-p} $year^{-1}$	Standard metabolism factor	Hartvig et al. 2011
p	0.75		West et al. 1997
ϵ	0.05	Efficiency of gonads	Hartvig et al. 2011
μ_0	1 g^{1-n} $year^{-1}$	Background mortality prefactor	Andersen and Pedersen 2010
ξ	0.1	Fraction of lipid reserves	
κ	8.2724×10^{6} $g^{(\lambda-1)}$ km^{-2}	Background spectrum carrying capacity	
λ	2.05	Resource exponent	Andersen and Beyer 2006
r_0	4 g^{1-p} $year^{-1}$	Resource growth rate	Savage et al. 2004
η_F	0.05	Fraction of W_∞ to recruit to trawl fishery	Andersen and Beyer 2013

$$f(w) = \frac{E(w)}{E(w) + \mathrm{h}w^n} \tag{A13}$$

where $\mathrm{h}w^n$ is the maximum consumption rate and n is a metabolic consumption exponent.

Feeding from the spectrum gives the individual in the model available energy that is divided between growth, reproduction, and respiration. The available energy is defined as

$$E_a(w) = \alpha f(w)\mathrm{h}w^n - k_s w^p \tag{A14}$$

where α is efficiency of energy conversion, k_s is a standard metabolism factor, and p is a standard metabolism exponent. The energy available after metabolism is divided between growth and reproduction as

$$\psi\left(\frac{w}{W_\infty}\right) = H\left(\frac{\eta_m W_\infty}{w}\right)\left(\frac{w}{W_\infty}\right)^{1-n} \tag{A15}$$

where η_m is the fraction of W_∞ that species start reproducing, and H is a smooth function described as $H(z) = (1 + z^{-10})^{-1}$. The energy available for somatic growth then becomes

$$g_i(w) = E_a(w)\left[1 - \psi\left(\frac{w}{W_{\infty_i}}\right)\right] \tag{A16}$$

The remaining energy after metabolism and growth is used for spawning with the ψ spawning rule. Spawning is counted as the total number of eggs, denoted as physiological reproduction, R_p:

$$R_{p,i} = \frac{\epsilon}{2w_0}\int_{w_0}^{W_{\infty_i}} E_a(w)\psi\left(\frac{w}{W_{\infty,i}}\right)N_i(w)\mathrm{d}w \tag{A17}$$

The recruitment of eggs into the population, R_i, is regulated by a Beverton–Holt like function. This function eliminates competitive exclusion and allows for co-existence of species.

$$R_i = R_{\max,i}\frac{R_{p,i}}{R_{p,i} + R_{\max,i}} \tag{A18}$$

where $R_{\max}$ is the maximum number of recruits. In the main paper, we use maximum recruitment to calibrate the two models and achieve similar relative biomasses between species in the size-based model and EwE.

The predation mortality (μ_p) that an individual experiences is directly related to the feeding presented earlier. It is described as

$$\mu_p(w_p) = \int_{w_0}^{\max(W_\infty)} \phi\left(\frac{w_p}{w}\right)[1 - f(w)]\gamma w^q \sum_i N_i(w)\mathrm{d}w \tag{A19}$$

Besides mortality due to predation, a background mortality is also imposed. This is to make sure that large individuals also experience some mortality, as they are exempt from predation mortality. Background mortality is assumed to scale with asymptotic size as

$$\mu_b = \mu_0 W_\infty^{n-1} \tag{A20}$$

Lastly, individuals can experience a starvation mortality (μ_s) if there is insufficient food. When food supply is too scarce, there is not negative growth, but rather just the added mortality

$$\mu_s(w) = \begin{cases} 0 & \alpha f(w)\mathrm{h}w^n > k_s w^p \\ \dfrac{k_s w^p - \alpha f(w)\mathrm{h}w^n}{\xi w} & \alpha f(w)\mathrm{h}w^n \le k_s w^p \end{cases} \tag{A21}$$

where ξ is the fraction of lipid reserves.

Early in life, fish are small larvae and therefore need other food than fish to grow. To fuel this part of the spectrum, there is a dynamic background spectrum symbolizing other parts of the community than fish. The background is modeled as continuous size classes ranging from 0.001 g to 50 g with the following equation:

$$\frac{\partial N_R(w)}{\mathrm{d}t} = r_0 w^{p-1}\left[\kappa w^\lambda - N_R(w)\right] - \mu_p(w)N_R(w) \tag{A22}$$

where r_0 is the growth rate, κ the carrying capacity, and λ is the slope of the spectrum. κ is used to get the total biomass in the system within the same order of magnitude as the EwE simulations.

The trawl selectivity in the size-based model is modeled as a smooth sigmoid function that changes from 0 to 1 as the weight of an individual approaches its asymptotic weight:

$$s = \left[1 + \left(\frac{w}{\eta_F W_{\infty_i}}\right)^{-10}\right]^{-1} \tag{A23}$$

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Appendix B

Biomass calibration

The calibration of the size-based model to EwE biomasses was performed in two steps. First, we found the carrying capacity of the background spectrum κ that best fit the total biomasses in the system under equilibrium. The fitting procedure was performed using the "fminbnd" function in MATLAB R2014b. This operation ensured that the total biomasses of the two systems were at the same magnitude. Secondly, we fitted the maximum recruitment parameter, R_{max}, for each species to achieve the same relative level of biomasses between species or groups (see Fig. 2B of the main text) using least sum of squares. The resulting values and residuals are listed in Table B1 and comparison of the system biomasses is provided in Fig. 1B (main text). The calibration gives the model similar baseline estimates of biomasses, so further exploration of fishing scenarios can be attributed to model properties and not parameterization. The calibration was performed with no fishing in either model. Running Ecosim to equilibrium with the fishing mortalities from Ecopath (Field et al. 2006) caused some species to go extinct, making equilibrium catches an unsuitable target for calibration.

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Table B1. Calibration parameters for the species in the size-based model.

Species[a]	R_{max} (1×10^6 eggs·year^{-1}·km^{-2})	Residual (EwE biomass – size-based biomass)	W_∞ (g)	Fishing type	Minimum catch size (g)
Mesopelagics	50	7.7616	13	Knife-edge	5
Benthic	19.181	3.9317	147	Knife-edge	5
Small flatfish	12.5614	3.4835	270	Knife-edge	10
Longspine thornyhead	6.8941	2.0652	272	Knife-edge	10
Pacific sardine	4.4228	1.3216	286	Knife-edge	5
Rex sole	1.105	0.3839	376	Knife-edge	10
Forage	50	25.582	483	Knife-edge	5
English sole	0.9836	0.3122	720	Knife-edge	10
Dover sole	0.6213	0.158	942	Knife-edge	10
Petrale sole	0.5263	0.0832	1 176	Knife-edge	10
Slope rockfish	0.026	0.0035	1 241	Knife-edge	10
Yellowtail rockfish	2.1376	−0.3563	1 702	Trawl	10
Pacific ocean perch	0.0498	−0.0113	1 860	Knife-edge	10
Pacific hake	18.7604	−9.0583	1 954	Trawl	50
Shelf rockfish	0.5339	−0.2199	2 108	Knife-edge	10
Mackerel	1.7236	−1.076	2 306	Knife-edge	5
Widow rockfish	0.3266	−0.2369	2 528	Trawl	50
Canary rockfish	0.0001	−0.0001	2 703	Trawl	10
Black rockfish	0.0565	−0.0753	3 265	Knife-edge	10
Arrowtooth flounder	0.1023	−0.2929	4 903	Knife-edge	10
Shortspine thornyhead	0.0025	−0.0103	6 096	Knife-edge	10
Macrourids	0.1223	−0.8438	8 847	Knife-edge	5
Sablefish	0.155	−1.1769	9 552	Trawl	10
Dogfish	0.1383	−1.3435	11 649	Knife-edge	10
Pacific salmon	0.0708	−0.8037	13 300	Knife-edge	10
Lingcod	0.041	−0.5151	14 961	Trawl	200
Albacore	0.0001	−0.1054	35 825	Trawl	200
Skates	0.0053	−0.2075	41 367	Trawl	200
Coastal sharks	0.0001	−0.1235	42 996	Trawl	50
Pacific halibut	0.0001	−0.3566	78 492	Trawl	200

[a]For species definitions, see Field et al. (2006).

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